MILITARY SPECIFICATION

COOLING-PUMPING GROUP

AN/AWA-6

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - Cooling-Pumping Group AN/AWA-6 specified herein shall accept three-phase, 400 cycle input power and provide portable hydraulic and electric power and cooling air through the normal aircraft external connections for powering and cooling the AMCS Aero 1A during F-level maintenance.

1.2 Classification - Cooling-Pumping Group AN/AWA-6 specified herein shall be of one type and shall consist of the following:

Unit	Type Designation	Appl. Para.
Truck, Dolly	V-181/AWA-6	3.4.1
Pumping Unit, Hydraulic	HD-416/U	3.4.2
Junction Box	J-1197/AWA-6	3.4.3
Cooler, Air, Electronic Equipment including Flexible Air Duct (MP1)	HD-415/U	3.4.4
Cable Assembly for Cooler, Air, Electronic Equipment	W1/AWM-8	3.4.5
Cable Assembly for Pumping Unit, Hydraulic	W2/AWM-8	3.4.6

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-3432	Cable, Flexible and Extra Flexible, 300 and 600 Volts
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 11)
MIL-F-5504	Filters and Filter Elements, Fluid Pressure, Hydraulic Micronic Type

FSC 4320

Source: https://assist.dla.mil -- Downloaded: 20260-19T14:01Z
Check the source to verify that this is the current version before use.

2.1	MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
	MIL-H-5440	Hydraulic Systems: Design, Installation and Tests of Aircraft
	MIL-I-6181	Interference Limits, Test and Design Requirements, Aircraft Electrical and Electronic Equipment
	MIL-M-7969	Motors, Alternating Current, 400-cycle 115/200 Volt System, Aircraft, General Specification
	MIL-N-18307	Nomenclature and Nameplate for Airborne Electronic and Associated Equipment
	MIL-T-17113	Tests; Shock, Vibration and Inclination (for Electronic Equipment), General Specification
	MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Specification for
	MIL-W-5088	Wiring, Aircraft, Installation of

STANDARDS

Military

MIL-STD-12	Abbreviations for use on Drawings
MIL-STD-103	Abbreviations (For Electrical and Electronic Use)
MIL-STD-108	Definitions of and Basic Requirements for Electrical and Electronic Equipment
MIL-STD-167	Mechanical Vibration of Shipboard Equipment
MIL-STD-200	Military Standard for Electron Tubes
MIL-STD-701	Transistors, Military Standard
AN 3114	Receptacle, External Power, 115/200V, 3 phase
AN 3430	Cable Assembly, Power, Electrical, 115/200 volts, 3 phase
MS 16051	Coupling, Ground Cooling, Combat Type Aircraft

Federal

FED-STD-595	Colors

PUBLICATIONS

Bureau of Naval Weapons

SAR-300	Material Changes and Material Bulletins, Preparation of

Source: https://assist.dla.mil -- Downloaded: 20261O-19T14:01Z
Check the source to verify that this is the current version before use.

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following factors shall govern:

(1) Parts and materials selected from the specifications listed in MIL-T-21200(ASG) or other Military procurement specifications approved by the procuring service.

(2) When the contractor can demonstrate that the use of standard MIL parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(3) When a non-standard part is used, prime consideration shall be given to meeting the requirements of Specification MIL-T-21200 with respect to general material characteristics (such as non-inflammability, fungus resistance, corrosion resistance, non-toxicity, etc.) and to meeting the environmental conditions specified therein as limited or extended by this specification.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall be first approved by the Bureau of Naval Weapons.

3.1.1 Nonstandard Part and Material Approval - Except for electron tubes, diodes and transistors, approval of nonstandard parts and materials, as defined by MIL-T-21200, shall be in the hands of the Bureau prior to the delivery of the equipment required by the contact. In lieu of data sheets (Figures 1 and 2 of MIL-T-21200), nonstandard parts approval requests shall accompany the submission of Preliminary List of Electronic Parts. This list shall conform with Figure 3 of MIL-T-21200 and shall suitably designate nonstandard parts. Data for nonstandard parts identification shall consist of the manufacturer's applicable purchase specifications or drawings.

3.1.2 Electron Devices - Tubes, Semiconductor Diodes, Transistors - The choice and application of electron devices in this catagory shall be made in accordance with the following requirements and in the sequence noted:

(1) MIL-STD-200 for electron tubes and semiconductor diodes

(2) MIL-STD-701 for Transistors

* (3) Guidance List of Semiconductor Diodes

* (4) List of Transistor Types

* For use in Navy equipment only.

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:01Z
Check the source to verify that this is the current version before use.

3.1.2.1 Electron Tube Complement - A tube complement report prepared on DD Form 816 shall be submitted to the Bureau of Naval Weapons for review and approval prior to preproduction testing. The tube complement shall include all electron tube types (including crystal diodes and transistors) used in the equipment. Types chosen from sources identified by (1) and (2) of paragraph 3.1.2 are preferred and are considered "standard". Types chosen from sources other than (1) or (2) are "nonstandard" and require specific procuring agency approval. To facilitate this approval, the tube complement report will be accompanied by a justification for each nonstandard electron device, a copy of the contractor's purchase specification and qualified sources of supply.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all requirements of Specification MIL-T-21200 applicable to Class 2B equipments for design, construction, and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of this equipment shall not exceed 275 pounds.

3.2.3 Operating Life

3.2.3.1 Operational Stability - When connected in the intended manner to an aircraft in which the AMCS Aero 1A is installed or to a test load* and operated in accordance with paragraph 3.2.25 this equipment shall operate with specified performance for 100 hours without the necessity for readjusting any controls which are inaccessible to the operator. Interruption of this operation and internal access to the 5 micron filter of Pumping Unit, Hydraulic HD-416/U shall be permissible during this period for filter replacement. Nonoperational controls which are accessible to the operator without equipment disassembly shall not require readjustment for a period of at least 24 hours.

3.2.3.2 Total Operating Life - With the exception of replaceable filters whose operating life is dependent on the degree of oil contamination and the motor-pump assembly, P/O HD-416/U, whose total operating life shall be at least 1000 hours, the equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 Reliable Operating Life - With replenishment of lubrication oil in the gear box, P/O HD-416/U, after each 100 hours of operation, the equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.4 Cabling and Connections - The equipment shall use cables and connectors in accordance with Specification MIL-T-21200. Reference symbols for connectors shall be chosen so that each item will have separate identity within an equipment. Wherever possible cable connectors shall be identified by marking the reference symbol either on the connector body or on a cable identification band near the connector. Suitable plastic markers shall be used for identification of cable and connector assemblies. The space adjacent to each external jack shall be marked with the applicable reference symbol. The marker for each cable connector which mates with a jack on this equipment shall also contain a parenthetical designation of the mating jack. When similar connectors are used they shall be keyed so as to avoid improper connection.

(1) External connections to units of this equipment shall be located to minimize interference of interconnecting facilities with operating controls and monitoring facilities of this equipment and need not be a front panel feature.

(2) The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet the interference control or test requirements, and provided the assembly of its plug may be easily accomplished.

* See Note 6.7 - Hydraulic Test Load Characteristics

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:01Z
Check the source to verify that this is the current version before use.

3.2.4.1 Interconnecting Cables - The equipment shall be capable of satisfactory operation using external wiring chosen from applicable specifications listed in MIL-T-21200 except in those situations where it is necessary to utilize special cabling to be compatible with the equipment under test. Wire size of individual conductors shall conform with current ratings established by MIL-W-5088.

3.2.4.2 Power Input Connections - Primary power input connections of the 3 ϕ, 115/200 V, 400 cps to this equipment shall be accomplished by a cable connector and control circuits which meet the intent of MS25245.

3.2.5 Control Panels - Insofar as is feasible, all operational controls, adjustments, overload protectors and indicators shall be located on a test set control panel. With the exception of multipin jacks (see paragraph 3.2.4), the surface visible to the operator adjacent to panel facilities shall be marked with a suitable word, phrase or abbreviation to indicate the function of the part. Where abbreviations are used, they shall be chosen from MIL-STD-12, MIL-STD-103 or as approved by the Bureau of Naval Weapons.

(1) With the exception of the pressure gauge, P/O HD-416/U, which shall have a white enamel dial and black scale markings with the operating pressure range and pointer finished with Nile green phosphorescent paint, all panel markings contained on the HD-415/U and HD-416/U shall be done in Nile green phosphorescent paint on a black background. The switch guards on the HD-415/U and HD-416/U shall have phosphorescent painted tips. When the design of a major unit includes front panel facilities, a proposed layout which delineates identifying titles, calibration marks and instrument scales shall be submitted to the Bureau of Naval Weapons for information and comment.

(2) The initial submission shall be accomplished before equipments are subjected to preproduction tests.

3.2.6 Interchangeability - The equipment shall meet the interchangeability requirements as specified in MIL-T-21200. Unless otherwise specified, interchangeability shall be measured against a model or drawings supplied by the Bureau of Naval Weapons.

3.2.7 Radio Interference Control - The equipment shall conform with the requirements of MIL-I-6181 with the following exceptions:

(1) In lieu of the CW radiated interference requirements of specification MIL-I-6181, Figure 18, no radio interference voltage in excess of 50 antenna induced microvolts shall radiate from this equipment over the frequency range of 150 KC to 400 mc.

(2) The requirements of paragraph 3.5 of MIL-I-6181 for conducted susceptibility shall be met by this equipment when the rf signal is limited to 1000 microvolts over the frequency range of 15 KC to 400 mc.

(3) The requirements of paragraph 3.5 of MIL-I-6181 for radiated susceptibility shall be met by this equipment when the frequency of the test signal is limited to the range of 15 KC to 400 mc.

3.2.8 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in paragraph 3.2.8 of MIL-T-21200 except that no set screw wrenches shall be mounted in the equipment.

3.2.9 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval shall be in accordance with MIL-N-18307.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:01Z
Check the source to verify that this is the current version before use.

3.2.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests on the equipment.

Warm Up Period	10 minutes
Temperature	Room Ambient (30°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1V, Line/Neutral, 3-phase
Input Power Frequency	400 ∓ 8 cps
* Oil Temperature	120 ∓ 5°F

3.2.11 Service Conditions - When connected in the intended manner to an aircraft in which AMCS Aero 1A is installed, this equipment shall operate in accordance with the operating duty cycle delineated by paragraph 3.2.25 to the extent specified herein under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2B equipments with exceptions as noted herein.

3.2.11.1 Vibration - In lieu of the requirements of MIL-T-21200, Paragraph 3.2.14.4, this equipment shall operate with specified performance when subjected to Type I environmental vibration as delineated by MIL-STD-167. Accessory mountings shall be loaded with the equipment they are designed to support or mock-up equivalents.

3.2.11.2 Temperature -

(1) In lieu of the requirements of MIL-T-21200, Paragraph 3.2.14.1, for intermittent operation at 71°C no operation shall be required above ambient temperatures of 55°C (131°F).

(2) Operation without damage, but not necessarily with specified performance, shall be required in ambient temperature ranges of 40°C (104°F) to 55°C (131°F) and -40°C (-40°F) to -20°C (-4°F).

(3) Operation with specified performance shall be required after a warm-up period of 10 minutes over an ambient temperature range of -20°C (-4°F) to +40°C (104°F).

3.2.11.3 Shock - In lieu of the requirements of MIL-T-21200 Paragraph 3.2.14.5.1, it shall be a design objective for the equipment as a whole, to withstand the Type II Class B high impact shock requirements of MIL-T-17113, in the vertical axis only, and subsequently operate with specified performance. Minor damage or distortion will be permitted providing such damage or distortion cannot in any way impair the ability of the equipment to perform its principal function.

3.2.12 Warm-Up Time - The warm-up time for the equipment shall not exceed ten minutes under standard conditions and at extreme service conditions. Warm-up time shall be exclusive of the purge cycle. (See paragraph 3.4.2.7.4). In the event of a primary power voltage loss of up to 10 seconds duration, which shall occur during normal operation, this equipment shall reattain normal operation per paragraph 3.3, without damage, within 1 minute after resumption of normal voltage.

3.2.13 Input Power Requirements - In lieu of the requirements of Specification MIL-T-21200, Paragraph 3.2.13.1.1 with respect to voltage levels and frequency ranges, the equipment shall give specified performance in accordance with paragraph 3.3 from the following power source when limits are as listed below. The power required shall not exceed the amounts specified.

(1) AC Power (three phase) . . . 115/200 volts, 400 cps
Wye connected

Operating Voltage Limits, Line to Neutral		108 to 121 volts
Operating Frequency Limits		380 to 420 cps
Reduced Performance Voltage Limits, Line to Neutral	Low:	102 to 108 volts
	High:	121 to 124 volts

* Applicable to MIL-H-5606 contained in Pumping Unit, Hydraulic HD-416/U.

Source: https://assist.dla.mil -- Downloaded: 202640-19T14:01Z
Check the source to verify that this is the currentversion before use.

3.2.13 Reduced Performance

Frequency Limits	Low:	350 to 380 cps
	High:	420 to 450 cps

(2) Total volt amperes (three phase)

Steady State @ 380 cps, 121 volts	25,385 VA max

3.2.13.1 Abnormal Voltage Protection - The equipment shall not be damaged by abnormal primary power voltage conditions and shall automatically resume normal operation when the voltage returns within limits. Abnormal voltage conditions shall not exceed the amplitudes and durations specified by Figures 1 and 2 of MIL-E-7894.

3.2.13.2 Overload Protection - Overload protection for the equipment shall be provided to the extent required by the detailed design section of this specification. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage to the equipment or shall have suitable protective devices. The use of fuses and other protective devices shall be held to a minimum.

3.2.13.3 Test Set, Grounds - In lieu of the requirements of MIL-T-21200 paragraphs 3.2.12.1 and 3.2.13.1.3(2) for operation of the equipment with an external ground on either side of the input power line, the operation of this equipment shall be limited to an ungrounded or a grounded neutral, 3 phase, 400 cps Wye connected power source.

(1) A common tie point shall be provided whereby enclosures of this equipment shall be interconnected to an electrical ground for the installation. This feature shall be accomplished by connecting the test equipment chassis ground(s) via the interconnection and/or power distribution cable(s) to a ground tie point provided by this equipment, or a unit thereof and a grounding cable assembly as delineated by paragraph 3.4.3.6 of this specification.

3.2.14 Standby Provisions - No standby provisions or space heaters shall be included in this equipment.

3.2.15 Moisture and Fungus-Resistant Treatment - Equipment shall be fungus-proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary.

3.2.16 Enclosures - Enclosures for this equipment shall meet the requirements of MIL-STD-108 to the extent delineated by the applicable detailed design paragraphs of this specification.

3.2.17 Transit Cases - No transit cases shall be furnished for the units of this equipment.

3.2.18 Mounting Racks - None.

3.2.19 Glass - Glass used in viewing windows of meter and pressure gauges furnished as part of this equipment shall be of the commercial quality normally supplied with such items and shall not require specific procuring agency approval.

3.2.20 Diagrams, Schematics and Instructions - No diagrams or schematics shall be attached to units or accessories of this equipment. Suitable caution and instruction plates shall be provided in the control panel cover of Pumping Unit, Hydraulic HD-416/U to insure safety of operating personnel and to prevent equipment misuse. These plates shall conform with paragraph 3.2.22 and be so located as to be conspicuously legible to the operator when he is facing the control panel of this unit in its "operate" condition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:01Z
Check the source to verify that this is the current version before use.

3.2.21 Marking and Identification - Marking and identification of this equipment shall be engraving, stamping, silk screening, decalcomania, labels, tags, or identification bands. Processes used shall be suitably protected so that the marking will remain legible throughout the life of the equipment. Marking shall not affect the performance of the equipment nor indicate the date the equipment was manufactured.

3.2.22 Painted Surfaces - In lieu of the requirements of MIL-T-21200, paragraphs 3.1.12.3, external surfaces of this equipment shall be primed with zinc chromate and finished with a yellow gloss enamel, color no. 13538 in Federal Standard 595. This enamel shall meet the requirements of Specification MIL-E-15090 enamel, Class I for weather-exposed surfaces except for color and the composition required to produce this color.

3.2.23 Part Identification - In accordance with the unit numbering of MIL-STD-16 each replaceable part shall be assigned a reference symbol by which it shall be identified in the equipment, schematic, instruction and maintenance manuals, and parts lists.

3.2.24 Pressure Requirements - Types of hydraulic components, delineated in Table I of MIL-H-5440, which are utilized in Pumping Unit, Hydraulic HD-416/U shall meet the operating, proof and burst requirements specified therein.

3.2.25 Operating Duty Cycle - When connected to an aircraft installed AMCS Aero 1A in the intended manner, or a test load*, and subjected to the appropriate warm-up period, Pumping Unit, Hydraulic HD-416/U shall operate with performance as specified herein when the duty cycle does not exceed 4 hours continuous operation with any one aircraft.

3.2.26 Power-On Indicator - No power-on indicator lamp as required by MIL-T-21200 Paragraph 3.2.13.1.3 shall be included in this equipment.

3.2.27 Audio Noise Requirements - When connected to an aircraft installed AMCS Aero 1A and operated at rated conditions in the intended manner, the AN/AWA-6 equipment shall not cause the sound level pressure, when measured at 5 feet from the equipment, to exceed 95 db above a reference level of 2×10^{-4} microbar (10^{-16} watts/cm^2). Such measurement shall be performed with sound level measuring equipment having a uniform frequency response over the range of 20 cps to 20 KC/S. Any noise deadening material used within this equipment shall not be adversely affected by hydraulic oil and shall not contribute as a fire hazard.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to the performance under both the standard and extreme service conditions. When reduced performance under extreme service conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified.

3.3.2 Test Facilities - The Cooling-Pumping Group AN/AWA-6 shall provide primary power, hydraulic power and cooling facilities for the major portions of the AMCS Aero 1A, in order that these portions may be operated and serviced while mounted in an aircraft.

3.3.3 Input Requirements

3.3.3.1 Primary Power Inputs - Power of the type listed in paragraph 3.2.13 shall be accepted from the ship's supply or other primary source and utilized to power this equipment, all associated equipment, and all subsidiary equipments.

(i) Junction Box J-1197/AWA-6 shall be capable of distributing 115/200V, 3 phase, 400 cps primary power as follows:

(a) Power for AMCS Aero 1A including:	5800 VA max
Power for AN/APQ-72	3600 VA max
Power for AN/AAA-4	200 VA max
Power for AN/APA-128	2000 VA max
(b) Power for Subsidiary Test Equipment including:	2185 VA max
Power for AN/APM-130	370 VA max
Power for Junction Box J-1197/AWA-6 Convenience Outlets	1815 VA max

* See Note 6.7 - Hydraulic Test Load Characteristics

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:01Z
Check the source to verify that this is the current version before use.

3.3.3.1	(c) Power for AN/AWA-6 including:	11,400 VA max
	Pumping Unit, Hydraulic HD-416/U	8300 VA max
	Cooler, Air, Electronic Equipment HD-415/U	3100 VA max
	(d) Power for A/C Facilities	6000 VA max

3.3.4 Output Requirements - This equipment shall provide the following outputs:

3.3.4.1 Electrical Outputs -

(1) This equipment shall be capable of distributing 115/200V, 3Ø, 400 cps primary power as follows:

(a) 11.8 KVA via the attached Junction Box cable assembly to the aircraft being serviced

(b) 2185 VA to Subsidiary Test Equipment

3.3.4.2 Hydraulic Output - This equipment, when operating under specified service conditions, warm-up period, and duty cycle shall provide pressurized filtered MIL-H-5606 hydraulic fluid to the aircraft being serviced having the following characteristics:

(1) With a return line pressure of 40 psi, nominal, the flow rate shall not be less than 2.5 gallons per minute at any output pressure, as measured at the unit pressure gauge, between 0 and 2400 psi. Fluid temperatures shall never exceed 107°C (225°F).

(2) The hydraulic fluid shall be delivered to the load through a built-in filter having a particle removal rating per MIL-F-5504 of 5 microns.

3.3.4.3 Cooling Air Output - This equipment shall provide forced cooling air to the aircraft being serviced having the following characteristics:

(1)	Air Temperature Rise between intake and fan outlet	No more than 15°F when intake air is 104°F
(2)	Capacity at discharge connection of Unit 4 output air housing	Not less than 459 cubic feet/minute, at an air density of 0.074 pounds/cubic foot, with a static pressure of 23 ± 1.6 inches of water gauge.
(3)	Filtration:	No more than 10% of dust particles greater than 0.01 millimeter in diameter present in the intake air shall appear in the fan output air.

3.3.5 Monitoring Facilities -

3.3.5.1 Pressure Gauge -

(1) Hydraulic pressure gauge shall be provided as part of Pumping Unit, Hydraulic HD-416/U and shall read the output hydraulic pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:01Z
Check the source to verify that this is the current version before use.

3.3.5.1 (a) Range - 0 to 5000 psi (marked in 100 psi steps)

(b) Accuracy - ± 2% of full scale indication

(c) Dial Size - 3-1/2 inch diameter

3.4 Detail Design Requirements

3.4.1 Truck, Dolly - The Truck, Dolly, V-181/AWA-6 shall meet the following requirements:

3.4.1.1 Function - The Truck, Dolly shall provide mounting and transportation means for the Pumping Unit, Hydraulic, and Cooler, Air, Electronic Equipment, and the Junction Box when used for F-level maintenance of the AMCS Aero 1A.

3.4.1.2 Form Factor and Mounting* - The Truck, Dolly shall be of handbarrow configuration suitable for one-man operation. This assembly shall include a welded tubular aluminum frame, reinforced by plates and gussets, with two pneumatic tired wheels at the forward end and two hand grips at the rear. The pneumatic type tires shall be 14 inches, nominal, in diameter and have a compliance which permits the loaded vehicle to pass over a 1-1/2 inch diameter cable without application of excessive operator force. Four rests with rubber feet shall be so arranged as to provide a horizontal position for the "in use" attitude of this vehicle. In this position the wheels shall not touch the floor or deck. In the "transport" position, with the hand grips elevated 27 inches, minimum, the forward rests shall have at least a two inch clearance above the floor. This vehicle shall include manually operated quick detach fastenings for mounting the Junction Box, J-1197/AWA-6 with its attached Pumping Unit, Hydraulic HD-416/U on the forward position (between the dolly wheels) of the Truck, Dolly with the control panels of both units facing forward. No tools shall be necessary to mount or detach J-1197/AWA-6 and HD-416/U from vehicle. The necessary brackets and shock mounts, whereby the Cooler, Air Electronic Equipment HD-416/U may be mounted at the rear of the Truck, Dolly, shall be provided as part of this vehicle. Means for mounting or detaching the Cooler may be accomplished by the use of small hand tools. The frame of the Truck, Dolly shall provide wrap-around stowage for the aircraft power cable and hydraulic hose assembly each of which are 18 feet in length. Means shall be provided to keep these stowed items clear of truck wheels and to retain the hydraulic hose fittings. In addition, four tie-down rings, compatible with standard aircraft tie-down devices, shall be securely attached to the vehicle frame. The overall dimensions of this unit shall not exceed 34 inches wide, 21-1/2 inches high or 64 inches long.

3.4.1.3 Weight - The weight of this unit shall not exceed 55 pounds.

3.4.1.4 Contents - As delineated by paragraph 3.4.1.2 above.

3.4.1.5 Controls - None.

3.4.1.6 Electrical Connections - None.

3.4.2 Pumping Unit, Hydraulic - The Pumping Unit, Hydraulic HD-416/U shall meet the following requirements:

3.4.2.1 Function - The Pumping Unit, Hydraulic, shall supply hydraulic power through the normal aircraft external hydraulic connections for powering the AS-940/APQ-72 antenna during F level maintenance on the AMCS Aero 1A.

3.4.2.2 Form Factor and Mounting - The Pumping Unit, Hydraulic HD-416/U shall, in its portage condition, be of rectangular parallel piped configuration with rounded edges and corners. The maximum dimensions exclusive of attached hoses, connectors, hinges, and strikes shall be 20 inches wide, 15 inches high and 14 inches deep. The allowable projection of louvres, strikes, latches, hinges, handles, etc. shall be 1/2 inch on either end of the case, 3/8 inch on the rear, none on the bottom, 1/4 inch on the top and 1/2 inch on the front. Electrical and hydraulic connectors may exceed this projection. This enclosure shall be dripproof, as defined by MIL-STD-108, in its portage and operate conditions. Removable louvred panels, secured by quick action fasteners, shall be furnished on the front, rear, and ends of the case to facilitate service access and provide ventilation. A recessed carrying handle shall be included on each end of the case. With handles extended with width dimension shall not exceed 26 inches. The upper front and forward top of the case shall form hinged, protective, "L" shaped cover for

* See Note 6.6 - Equipment Orientation

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:01Z
Check the source to verify that this is the current version before use.

3.4.2.2

an inclined control panel. In the "operate" condition this cover shall fold back on the top of the case to provide access to the panel facilities thereby increasing the height dimension of this unit to 19 inches maximum and the depth, to 16 inches maximum. The base of the case shall contain bosses for mounting alignment; include strikes to accept external fastenings; serve as a drip pan to collect approximately one gallon of oil leakage; and be equipped with a drain tube assembly to facilitate ultimate disposal of leaked oil. This unit shall mount on the top of the Junction Box J-1197/AWA-6 picking up locating pins thereon and being retained by latches located on the front, rear, and ends thereof. The front of the pumping unit shall be the 20" x 15" face of the equipment and so located as to be easily accessible from the operator's "in use" position (i.e., Standing in front of the cart). Facilities on the front face shall include semipermanently attached high and low pressure hydraulic oil hoses; purge tank connections, drain tube assembly for the unit base and the inclined control panel which mounts the purge tank filler assembly, pressure gauge, circuit breaker, and motor control switches.

3.4.2.3 Weight - The weight of this unit including oil and hoses shall not exceed 115 pounds.

3.4.2.4 Contents - This unit shall consist of an electrically driven variable displacement pump, a high pressure filter, pressure relief valve, pressure gauge, high pressure hose, return line hose, low pressure manifold, thermal expansion valve and pump residue filter with interconnecting hydraulic piping and fittings as required. In addition, a vented purge tank (with drain and high and low pressure connections), liquid level sight gauge, filler port assembly (including strainer and sealer cap), shall be a separate entity which is only connected to the previously described pumping system by the hydraulic hoses for purging entrapped air therefrom prior to its connection to the aircraft hydraulic facilities. The electrical system shall include a power connector, circuit breaker, motor control circuit consisting of a relay, flow switch, push-to-start and push-to-stop switches and the pump drive motor with reduction gear box. The characteristics of these components when operating under specified standard conditions and duty cycle using the aircraft hydraulic system as a load (a hydraulic load per 6.7 may be substituted for bench tests) shall be as follows:

(1) Variable displacement, motor driven pump designed for use with MIL-H-5606 hydraulic fluid capable of providing pressure and flow rates consistent with applicable requirements of paragraph 3.4.2.7.1 when energized by primary power described by paragraph 3.2.3.

(2) System filter, manually replaceable, metal mesh type, capable of providing filtration consistent with the applicable requirements of paragraph 3.4.2.7.1(3).

(3) A pressure relief valve, shall be provided to bypass oil to the return line manifold when the system pressure is excessive. The maximum differential rated flow pressure shall be factory adjusted to 3200 psi; the minimum reseat pressure shall be 2740 psi.

(4) Unit output pressure gauge reading 0-5000 ± 2 percent of full scale on a 3-1/2" diameter dial with indications every 100 psi.

(5) Hose Assembly - The assembly shall include two 18 ± 0.5 ft hoses one end of each semipermanently attached to HD-4167U. Anti-kink devices shall be included at the attached end for a length of one foot and the assembly shall be banded at two foot intervals. The pressure line (0.438 inches inside diameter, 0.719 inches outside diameter, nominal) and return line (0.625 inches inside diameter, 0.797 inches outside diameter, nominal) shall be teflon lined steel braid covered hose with an oil and electrical resistant covering. Self sealing quick disconnect fittings shall terminate the free end of the assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:01Z
Check the source to verify that this is the current version before use.

3.4.2.4 (6) Return Line Pressure Relief Valve - The assembly shall release oil from the return line manifold to the case base drip pan when the oil pressure due to thermal expansion becomes excessive. This valve shall be factory adjusted for a 2 gpm full flow pressure of 90 psi and have a minimum reseat pressure of 72 psi. The full flow pressure setting under service conditions using MIL-H-5606 fluid and during valve life shall not exceed ± 10%.

(7) Pump Residue Oil Bypass Filter - This assembly shall have a 10 micron rating per MIL-F-5504 and a capacity of 0.5 gallons per minute.

(8) Purge Tank Assembly- This assembly shall be constructed of stainless steel, be vented to the atmosphere and have a capacity of not less than one gallon. Quick disconnect fittings connected to the bottom of the tank shall be provided to accept the return pressure hoses. The tank construction shall preclude oil spillage at any attitude and include a fill opening with strainer and sealer cap. The tank shall not be connected to the hydraulic system except by pressure and return line hose attachment for the purging operation. A sight level gauge shall be included to indicate the safe liquid level operating range and to permit air bubble observation. A drain valve by which the purge tank may be emptied shall be included.

(9) Electrical Connector - The connector shall be provided to accept the input power and shall be located on the left side of the unit.

(10) Electrical Circuit Breaker - The electrical circuit breaker shall be a 3-pole manual reset type having a momentary current rating of 110 amperes for 6.5 + 3.5 seconds and a maximum continuous rating of 33 amperes. This item shall be suitable for 115/200V, 3φ, 400 cps operation.

(11) Motor Control Circuit - The motor control circuit shall consist of a push-to-start switch, push-to-stop switch, flow switch and power control relay, connected between the load side of the circuit breaker and the electrical input to the pump drive motor. The circuit shall be operative only when the main power control switch located on Junction Box J-1197/AWA-6 is in the "ON" position. In event of primary power interruption for more than 60 milliseconds or an oil flow rate of less than .5 gpm, this circuit shall de-energize the pump motor.

(12) Internal Connections - All tubing and fittings shall be stainless steel except the quick-disconnect fittings, which may be anodized aluminum. The pressure line shall have an outside diameter of 0.375 inches, nominal, and an inside diameter of 0.340 inches, nominal. The return line shall have an outside diameter of 0.750 inches, nominal, and and inside diameter of 0.695 inches nominal. All lines connecting to the pump shall be teflon-lined stainless steel braid-covered hose.

3.4.2.5 Controls - Front Panel Controls for the equipment shall include the circuit breaker and push type switches for starting and stopping the pump motor.

3.4.2.6 Connections - The electrical connections to HD-416/U shall be accomplished by a receptacle located on the left* end of the case near the top and rear. Hydraulic connections shall include the high pressure and return line hose assembly attached to the left* front of case and purge tank connections on the right*.

3.4.2.7 Operational Requirements - This unit, when operating under specified standard conditions, warm-up period, duty cycle and equipped with a clean output filter, using the aircraft hydraulic system as a load (a load may be substituted for test purposes**), shall meet the following requirements:

* See Note 6.6 - Equipment Orientation

** See Note 6.7 - Hydraulic Test Load Characteristics

Source: https://assist.dla.mil -- Downloaded: 202640-19T14:01Z
Check the source to verify that this is the current version before use.

3.4.2.7.1 Capacity - The unit shall circulate filtered, pressurized MIL-H-5606 hydraulic fluid having the following characteristics:

(1) When the return line pressure is 40 psi nominal, the flow rate shall not be less than 3 gallons per minute at any output pressure as indicated at the unit pressure gauge, between 0 and 2550 psi.

(2) For bench test purposes with 40 psi nominal pressure in the return line, the output pressure as indicated at the unit pressure gauge shall not exceed 3100 psi at zero flow rate.

(3) The hydraulic fluid shall be delivered to the load through a built-in filter having a particle removal rating per MIL-F-5504 of 5 microns.

3.4.2.7.2 Main Power Control - The unit shall be wired so that it will operate only when the main power control switch in the Junction Box is ON.

3.4.2.7.3 Purge Cycle - It shall be permissible to subject this unit to a suitable purge cycle, accomplished in accordance with maintenance handbook requirements, to prevent contamination of the aircraft hydraulic system with air from this unit.

3.4.2.8 General Requirements -

3.4.2.8.1 Panel Markings - All connectors, switches, gauges, hydraulic outlets, etc. shall be plainly marked. All marking shall be done in Nile green fluorescent paint on a black background. A warning sign shall be located on the control panel, reading as follows:

"DANGER: DO NOT START UNTIL PERSONNEL ARE CLEAR OF AIRCRAFT AND ANTENNA"

3.4.3 Junction Box - The Junction Box J-1197/AWA-6 shall meet the following requirements:

3.4.3.1 Function - The Junction Box shall accept 115 volt, 3-phase, 400 cycle input power and distribute same to the Pumping Unit, Hydraulic, and to the Cooler, Air, Electronic Equipment, and the aircraft under F-level test. It shall also provide five outlets for test equipment and a main power control switch. The main power control switch, by interrupting the control circuit, shall control power to the Pumping Unit, Hydraulic, and aircraft while the Cooler, Air, Electronic Equipment, shall continue to operate.

3.4.3.2 Form Factor and Mounting* - The Junction Box J-1197/AWA-6 shall be of box configuration with rounded edges and corners. The maximum dimensions exclusive of attached cable, controls, connectors, locating pins and latches shall not exceed 20 inches in width, 14 inches in depth and 6 inches in height. The enclosure shall be drip proof as defined by MIL-STD-108 in its portage and operate conditions. A removable panel in the base of this enclosure shall be included to provide access for wiring and servicing this unit. As mounted on the Truck, Dolly V-181/AWA-6 the front of the Junction Box shall be the 20" x 6" face of the equipment which faces forward in the transport attitude. This face shall include the attached primary power output cable assembly, ground stud, with grounding cable, AN/APM-130 output power receptacle and power switch with safety guard. The right end of the enclosure shall mount an input primary power receptacle. The rear face shall feature power output receptacles for the Cooler, Air Electronic Equipment, the Pumping Unit Hydraulic, active and spare fuses for the line to neutral connected receptacles which are located on the left end of the enclosure. The Pumping Unit, Hydraulic, which mounts on the top face of the Junction Box, shall be positioned by four locating pins and fastened by four manually operated latches which are included as Junction Box facilities.

* See Note 6.6 - Equipment Orientation.

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:01Z
Check the source to verify that this is the current version before use.

3.4.3.3 Weight - The weight of this unit including the attached output cable shall not exceed 45 pounds.

3.4.3.4 Contents - In addition to mechanical features identified by paragraph 3.4.3.2, power control facilities covered by paragraph 3.4.3.5, and electrical connections detailed by paragraph 3.4.3.6, this unit shall contain a means for interconnecting the chassis, or frame of the AN/AWA-6 units to an external ground. These provisions shall include a 3/8 inch ground stud with suitable hardware, electrically bonded to the Junction Box front face, to which all unit grounds are internally connected, and a cable assembly as required by paragraph 3.4.3.6(6). Electrical interconnections between the connectors of this unit shall be suitable for primary power distribution.

3.4.3.5 Controls - When interconnections between primary power, the AN/AWA-6 and the aircraft system under test are completed, means shall be included in this unit whereby the Cooler, Air Electronic Equipment together with any accessories connected to test equipment outlets of the Junction Box shall receive primary power controlled by their individual off on switches, whereas the power facilities to the Pumping Unit, Hydraulic and aircraft system shall be energized only when the power control switch on the Junction Box is in the "ON" position.

3.4.3.6 Electrical Connections - The following electrical connections shall be included:

(1) The primary power input receptacle shall be a recessed AN 3114-1B six pin male connector.

(2) Receptacles for line (phase A) to neutral primary power distribution to auxiliary test equipment shall include a weather-proof, two blade with ground pin, duplex receptacle and a receptacle per Raytheon Specification S276MS611 P029 with chain and cap.

(3) The receptacle to which the Cooler, Air Electronic Equipment is interconnected by W-1 cable assembly.

(4) The receptacle to which the Pumping Unit, Hydraulic is interconnected by W-2 cable assembly.

(5) The power cable to the aircraft shall be similar to AN 3430 except that it shall have four no. 4 AWG wires for phase A, B, C and neutral of the 3 phase 400 cps 115/200 Volt power and two no. 12 AWG wires for control purposes. One end of this cable assembly shall be permanently attached to the Junction Box through an elbow and clamp which facilitates stowing the cable by coiling it in a counterclockwise direction about the Junction Box as viewed from the top. The length of this cable shall be 216 ± 4 inches.

(6) The grounding cable assembly shall consist of a 144 ± 1 inch length of no. 4 AWG gauge flexible cable one end of which is terminated with a spade lug (compatible with the ground stud) the other with a 200 ampere battery clip.

3.4.3.7 Operational Requirements - The Junction Box shall be capable of providing electrical power outputs delineated in paragraph 3.3.4.1 when furnished with the required input power.

3.4.3.8 General Requirements -

3.4.3.8.1 Marking Requirements - Each receptacle, fuse, switch, etc. shall be identified. Methods and processes shall be in accordance with paragraphs 3.2.21 and 3.2.23.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T14:01Z
Check the source to verify that this is the current version before use.

3.4.4 Cooler, Air, Electronic Equipment - The Cooler, Air, Electronic Equipment, HD-415/U shall meet the following requirements:

3.4.4.1 Function - The Cooler, Air, Electronic Equipment, shall supply cooling air through the normal ground cooling connection to the aircraft for cooling the AMCS Aero 1A during F-level maintenance operations.

3.4.4.2 Form Factor and Mounting - This equipment shall consist of a base, intermediate housing and output air assemblies, which are generally of cylindrical configuration, and concentrically assembled along the cylinder axis. This equipment shall be capable of operation when the cylinder axis is either in a vertical or horizontal position. When the cylinder axis is in a vertical position, with air hose attached, the equipment enclosure shall be dripproof as defined by MIL-STD-108.

3.4.4.2.1 Size - The unit shall not exceed the following dimensions:

(1)	Base diameter, including protrusion of filter rain shield, mounting feet and carrying handle -	18 in.
(2)	Base diameter of noise/filter assembly -	14-1/8 in.
(3)	Height of noise/filter assembly -	8 in.
(4)	Diameter of intermediate external housing, including any flanges but excluding housing for electrical facilities -	7-7/8 in.
(5)	Diameter of air output facility -	5 in.
(6)	Overall height -	25 in.

3.4.4.3 Weight - The unit shall not exceed 50 pounds including air duct but exclusive of power cable.

3.4.4.4 Contents of Unit - The Cooler, Air, Electronic Equipment shall contain the following:

3.4.4.4.1 Base - The subassembly shall be a cylindrical compartment lined with sound absorbent material and a removable dust filter with a screen guard for mechanical protection. Three pads shall be permanently mounted to the periphery of the base at approximately 120-degree intervals for use as feet. The base top shall have a mounting flange to accept the intermediate housing assembly.

3.4.4.4.2 Intermediate Housing - The subassembly shall include:

(1) A 115/200 Volt, 400 cps, 3 phase electric motor in accordance with Specification MIL-M-7969 and compatible with the performance and primary power requirements of this specification. The motor enclosure shall be of such configuration as to preclude the ignition of an ambient explosive gaseous mixture with air where the principal contaminant is combat aircraft fuel vapor.

(2) A cylindrical housing assembly compatible with the base assembly and output air housing.

(3) A suitable carrying handle.

(4) A housing for electrical facilities. This housing shall be positioned 90 degrees from the carrying handle and centrally located on the periphery of the cylindrical intermediate housing assembly above the rain/splash filter shield. This housing shall contain the circuit breaker on-off switch and provide mounting for the receptacle.

(5) A removable rain/splash filter shield.

Source: https://assist.dla.mil -- Downloaded: 202640-19T14:01Z
Check the source to verify that this is the currentversion before use.

3.4.4.4.3 Output Air Housing - The subassembly shall be of metallic construction and compatible with the intermediate housing assembly and output air facility. Additional noise suppressing material may be contained in this unit.

3.4.4.4.4 Flexible Air Duct (MP-1) - The flexible duct shall be 12 + 0.5 feet long and 5 inches, nominal, in diameter. One end shall have a semi-rigid cuff capable of mating with the output air facility of the Cooler, Air, Electronic Equipment. This cuff shall incorporate a quick-detachable clamp. The other end of the duct shall have a transition piece from a 5 to 4 inch diameter and terminate in an MS 16051 coupling. The weight of this item shall not exceed 10 pounds.

3.4.4.5 Controls -

(1) Electrical Circuit Breaker - The circuit breaker shall be a 3-pole manual reset type having a momentary current rating of 42 amperes for 2 + 1 seconds and a maximum continuous rating of 11 amperes. This item shall be suitable for 115/200V, 3 phase, 400 cps operation.

3.4.4.6 Electrical Connections - The electrical connection to this unit shall be of an approved military type.

3.4.4.7 Requirements - The Cooler, Air, Electronic Equipment, shall meet the following requirements:

3.4.4.7.1 Discharge Temperature - The unit shall be so designed that the discharge air temperature at the fan outlet shall not exceed 119 degrees F. when the input air is 104 degrees F. maximum.

3.4.4.7.2 Flow Rate and Pressure - Under conditions of Paragraph 3.4.4.7.1 above, the flow rate shall be a minimum of 459 cubic feet per minute into a water gauge pressure of 23.0 ± 1.6 inches. (Air density equals 0.074 lbs/cu.ft.).

3.4.4.7.3 Filtration - The inlet air filter shall be capable of arresting 90 percent of dust particles exceeding 0.01 millimeters in diameter from the intake air.

3.4.5 Cable Assembly, Cooler, Air, Electronic Equipment (W-1) - The W1 cable shall be 72 + 3/4 inches long. One end shall have a connector (W1P1) capable of being connected to the Cooler, Air, Electronic Equipment (4J1), and the other end shall have a connector (W1P2) capable of being connected to the Cooler, Air, Electronic Equipment outlet (3J3) on the rear of the Junction Box. The cable shall be as specified by MIL-C-3432. Wire size and pin connections are to conform with unit wiring.

3.4.6 Cable Assembly, Pumping Unit, Hydraulic (W-2) - The W2 cable shall be 24 + 3/4 inches long. One end shall have a connector (W2P1) capable of being connected to the Pumping Unit, Hydraulic, and the other end shall have a connector (W2P2) capable of being connected to the Pumping Unit, Hydraulic, outlet (3J5) on the rear of the Junction Box. The cable shall be as specified by MIL-C-3432. Wire size and pin connections are to conform with unit wiring.

3.5 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of this equipment.

3.5.1 Complementary Equipment - It is recommended that the equipment covered by this specification be complemented by the following items and facilities which are not furnished as part of this equipment:

Source: https://assist.dla.mil -- Downloaded: 20260-19T14:01Z
Check the source to verify that this is the current version before use.

3.5.1 (1) A suitable cable assembly*, such as Type AN-3430, meeting the following requirements:

(a) The cable length shall permit placement of this equipment adjacent to the aircraft being serviced;

(b) The connector, which mates with this equipment, shall be compatible with a Type AN 3114-1B jack;

(c) The number of conductors shall be sufficient to interconnect the primary power source and control facilities to this equipment.

(2) Hydraulic Fluid per MIL-H-5606

*(The cable assembly for this application shall be the same as that used for applying primary electrical power to the aircraft being serviced when this equipment is not used).

Source: https://assist.dla.mil -- Downloaded: 202640-19T14:01Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, MIL-T-17113 and MIL-STD-167 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the Government Inspector after the equipment has successfully passed all individual tests.

Source: https://assist.dla.mil -- Downloaded: 202640-19T14:01Z
Check the source to verify that this is the current version before use.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 8 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3g's

Source: https://assist.dla.mil -- Downloaded: 20260-19T14:01Z
Check the source to verify that this is the current version before use.

4.4.1.3 The equipment shall be vibrated for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in 4.4.1.3, except that the test duration shall be 30 hours.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity).

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:01Z
Check the source to verify that this is the current version before use.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipment selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case of design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following: When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests).

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room Ambient
A.C. voltage	115 ± 5 volts (at lowest applicable frequency)

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilise. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:01Z
Check the source to verify that this is the current version before use.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon comple tion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.9 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-900.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:01Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment in the contract or order in accordance with Specification MIL-P-7936(AER).

6. NOTES

6.1 Use - This equipment shall be used to facilitate "F" level maintenance of the AMCS Aero 1A.

6.2 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 115 ± 1 volt AC line to neutral. This voltage value is to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.9. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.5 Performance Objectives- Minimum size and weight, simplicity of maintenance and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. The use of materials, parts and processes other than those required by Specification MIL-T-21200 shall be investigated and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability can be gained by their use, a request for approval of a non-standard part shall be submitted to the Bureau of Naval Weapons for consideration. Each request shall be accompanied by complete supporting information.

6.6 Equipment Orientation - Unless otherwise stated herein, position and direction as applied to the AN/AWA-6 shall be established in the transport attitude of the equipment as pushed by the operator. Thus the forward or front position is the wheel end of the Truck, Dolly V-181/AWA-6; the rear is at the hand grip end; the right side of any panel or unit is on the operator's right.

6.7 Hydraulic Test Load Characteristics - For bench test purposes a hydraulic test load shall consist of a suitable throttle valve, and a closed oil/air reservoir. Means for measuring pressure, flow rate and oil temperature may be included. The load shall have the following capabilities:

Source: https://assist.dla.mil -- Downloaded: 20260-19T14:01Z
Check the source to verify that this is the current version before use.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

6.7	(1)	Maximum working pressure	3200 psi
	(2)	Maximum flow rate	5 gal/min
	(3)	Reservoir capacity	13 ± 1 gal
	(4)	Return line pressure	40 psi nominal
	(5)	Heat Exchanger requirements:	
		The heat exchanger shall be capable of maintaining the oil temperature, measured at the reservoir intake, as follows:	
		(a) For standard conditions	49° ± 2.78°C (120° ± 5°F)
		(b) For extreme high temperature service conditions	107°C max (225°F)

Source: https://assist.dla.mil -- Downloaded: 202640-19T14:01Z
Check the source to verify that this is the currentversion before use.